

ARTIFICIAL INTELLIGENCE

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COURSE CODE:1708

1. Write the python program to solve 8-Puzzle problem

Python code:

```
from collections import deque
```

```
s=input("Enter puzzle: ")
```

```
q=deque([(s,[])])
```

```
v={s}
```

```
while q:
```

```
    x,p=q.popleft()
```

```
    if x=="123456780":
```

```
        print("Solution:")
```

```
        for a in p+[x]:
```

```
            print(a[:3]);print(a[3:6]);print(a[6:]);print()
```

```
            break
```

```
        i=x.index("0")
```

```
        for j in [i-3,i+3,i-1,i+1]:
```

```
            if 0<=j<9 and abs(i%3-j%3)<=1:
```

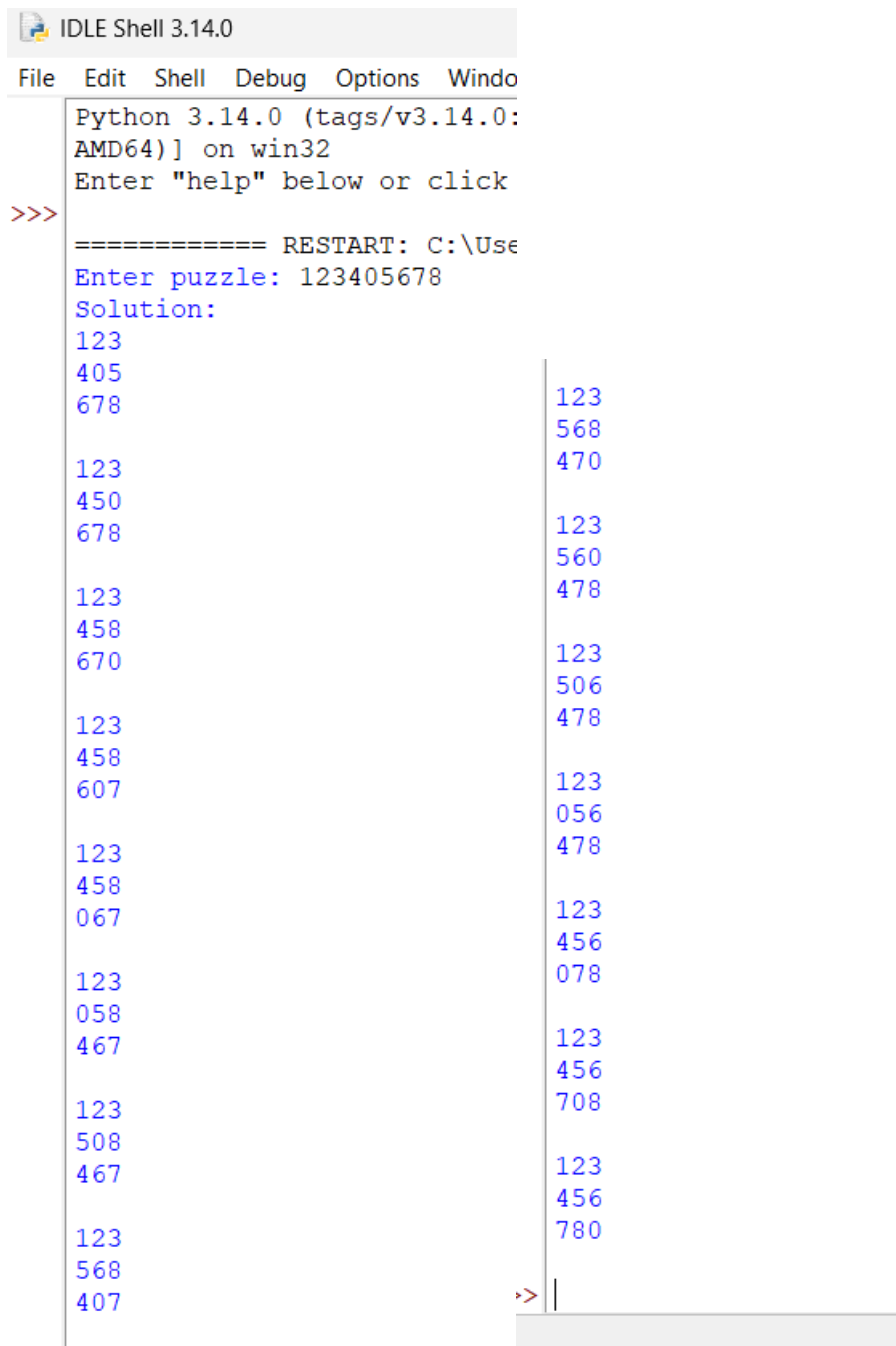
```
                a=list(x);a[i],a[j]=a[j],a[i];a="".join(a)
```

```
                if a not in v:v.add(a);q.append((a,p+[x]))
```

else:

print("No solution")

Output:



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window
Python 3.14.0 (tags/v3.14.0:
AMD64) on win32
Enter "help" below or click
>>>
===== RESTART: C:\Use
Enter puzzle: 123405678
Solution:
123
405
678
123
450
678
123
458
670
123
458
607
123
458
067
123
058
467
123
508
467
123
568
407
123
568
470
123
560
478
123
506
478
123
056
478
123
456
078
123
456
708
123
456
780
>>
```

2. Write the python program to solve 8-Queen problem

Python code:

```
def solve(r, q):

    if r == 8:

        for i in range(8):

            print(" ".join("Q" if q[i] == j else "." for j in range(8)))

        return True

    for c in range(8):

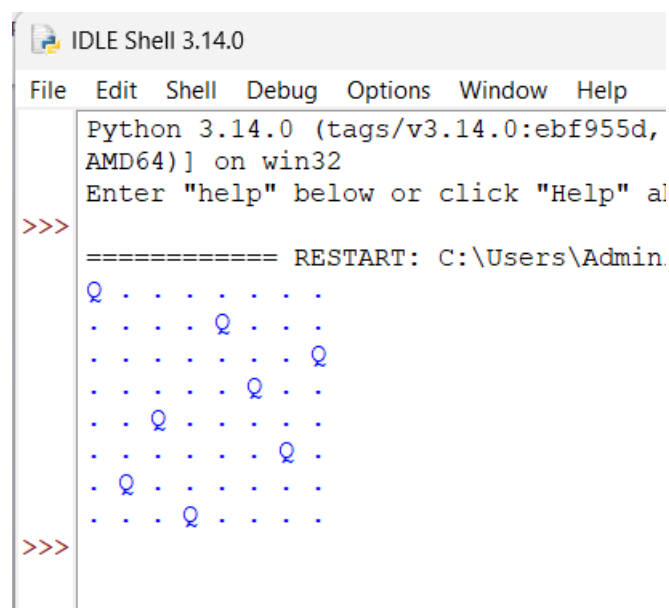
        if c not in q and all(abs(c-x) != r-i for i,x in enumerate(q)):

            if solve(r+1, q+[c]):

                return True

    return False
```

```
solve(0, [])
```

Output:

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, AMD64) on win32
Enter "help" below or click "Help" al
>>>
===== RESTART: C:\Users\Admin.
Q . . . . .
. . . Q . .
. . . . . Q
. . . . Q .
. . Q . . .
. . . . . Q
. Q . . . .
. . . Q . .
>>>
```

3. Write the python program for Water Jug Problem

Python code:

```
a = 0
```

```
b = 0
```

```
print(a, b)
```

```
while a != 2:
```

```
    if a == 0:
```

```
        a = 4
```

```
    elif b == 3:
```

```
        b = 0
```

```
    else:
```

```
        x = min(a, 3-b)
```

```
        a = a-x
```

```
        b = b+x
```

```
print(a, b)
```

Output:

```
IDLE Shell 3.14.0
File Edit Shell Debug Options
Python 3.14.0 (tags/v:
AMD64) ] on win32
Enter "help" below or
>>>
===== RESTART:
0 0
4 0
1 3
1 0
0 1
4 1
2 3
>>>
```

4. Write the python program for Missionaries Cannibal problem

Python code:

```
from collections import deque
```

```
q = deque([(3,3,1), []])
```

```
v = {(3,3,1)}
```

```
moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
```

```
while q:
```

```
    (m,c,b), p = q.popleft()
```

```
    if (m,c,b) == (0,0,0):
```

```
        for x in p + [(m,c,b)]:
```

```
            print(x)
```

```
        break
```

```
    for dm,dc in moves:
```

```
        nm = m-dm if b else m+dm
```

```
        nc = c-dc if b else c+dc
```

```
nb = 1-b
```

```
if 0<=nm<=3 and 0<=nc<=3:
```

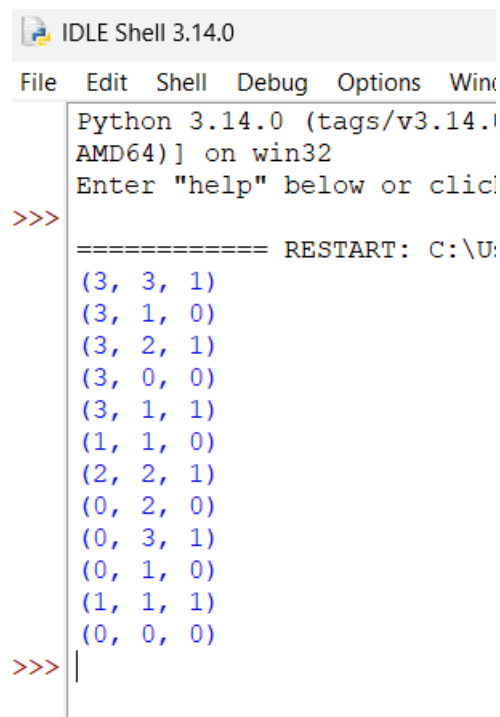
```
    if (nm==0 or nm>=nc) and (3-nm==0 or 3-nm>=3-nc):
```

```
        s=(nm,nc,nb)
```

```
        if s not in v:
```

```
            v.add(s); q.append((s,p+[(m,c,b)]))
```

Output:



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window
Python 3.14.0 (tags/v3.14.0) on win32 AMD64] Enter "help" below or click
>>>
===== RESTART: C:\U:
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
>>> |
```

5. Write the python to implement Travelling Salesman Problem

Python code:

```
from itertools import permutations
```

```
d=[[0,10,15,20],[10,0,35,25],[15,35,0,30],[20,25,30,0]]
```

```
best=999
```

```
path=None
```

```
for p in permutations(range(1,4)):
```

```
    r=(0,)+p+(0,)
```

```
    cost=sum(d[r[i]][r[i+1]] for i in range(4))
```

```
    if cost<best:
```

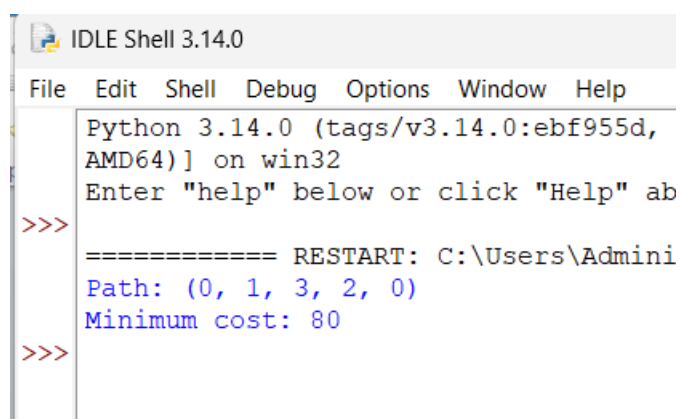
```
        best=cost
```

```
        path=r
```

```
print("Path:",path)
```

```
print("Minimum cost:",best)
```

Output:



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, AMD64) on win32
Enter "help" below or click "Help" ab
>>>
===== RESTART: C:\Users\Admini
Path: (0, 1, 3, 2, 0)
Minimum cost: 80
>>>
```